

The Roles of Caloric Restriction Mimetics in Central Nervous System Demyelination and Remyelination.

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The dysfunction of myelinating glial cells, the oligodendrocytes, within the central nervous system (CNS) can result in the disruption of myelin, the lipid-rich multi-layered membrane structure that surrounds most vertebrate axons. This leads to axonal degeneration and motor/cognitive impairments. In response to demyelination in the CNS, the formation of new myelin sheaths occurs through the homeostatic process of remyelination, facilitated by the differentiation of newly formed oligodendrocytes. Apart from oligodendrocytes, the two other main glial cell types of the CNS, microglia and astrocytes, play a pivotal role in remyelination. Following a demyelination insult, microglia can phagocytose myelin debris, thus permitting remyelination, while the developing neuroinflammation in the demyelinated region triggers the activation of astrocytes. Modulating the profile of glial cells can enhance the likelihood of successful remyelination. In this context, recent studies have implicated autophagy as a pivotal pathway in glial cells, playing a significant role in both their maturation and the maintenance of myelin. In this Review, we examine the role of substances capable of modulating the autophagic machinery within the myelinating glial cells of the CNS. Such substances, called caloric restriction mimetics, have been shown to decelerate the aging process by mitigating age-related ailments, with their mechanisms of action intricately linked to the induction of autophagic processes.

Oligodendrocytes in Huntington's Disease: A Review of Oligodendrocyte Pathology and Current Cell Reprogramming Approaches for Oligodendrocyte Modelling of Huntington's Disease.

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Huntington's disease (HD) is an autosomal dominant neurodegenerative disorder traditionally characterized by the selective loss of medium spiny neurons in the basal ganglia. However, it has become apparent that white matter injury and oligodendrocyte dysfunction precede the degeneration of medium spiny neurons, garnering interest as a key pathogenic mechanism of HD. Oligodendrocytes are glial cells found within the central nervous system involved in the production of myelin and the myelination of axons. Myelin is a lipid-rich sheath that wraps around axons, facilitating signal conduction and neuronal viability. The degeneration of myelin hinders effective communication and leaves neurons vulnerable to external damage and subsequent degeneration. Abnormalities in oligodendrocyte maturation have been established in the HD human brain, however, investigations into the underlying dysfunction of human oligodendrocytes in HD are limited. This review will detail the involvement of oligodendrocytes and white matter damage in HD. Recent developments in modeling human-specific oligodendrocyte pathology in HD will be discussed, with a particular focus on emerging somatic cell reprogramming approaches.

Cspg4 sculpts oligodendrocyte precursor cell morphology.

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The extracellular matrix (ECM) provides critical biochemical and structural cues that regulate neural

development. Chondroitin sulfate proteoglycans (CSPGs), a major ECM component, have been implicated in modulating oligodendrocyte precursor cell (OPC) proliferation, migration, and maturation, but their specific roles in oligodendrocyte lineage cell (OLC) development and myelination in vivo remain poorly understood. Here, we use zebrafish as a model system to investigate the spatiotemporal dynamics of ECM deposition and CSPG localization during central nervous system (CNS) development, with a focus on their relationship to OLCs. We demonstrate that ECM components, including CSPGs, are dynamically expressed in distinct spatiotemporal patterns coinciding with OLC development and myelination. We found that zebrafish lacking *cspg4* function produced normal numbers of OLCs, which appeared to undergo proper differentiation. However, OPC morphology in mutant larvae was aberrant. Nevertheless, the number and length of myelin sheaths produced by mature oligodendrocytes were unaffected. These data indicate that *Cspg4* regulates OPC morphogenesis in vivo, supporting the role of the ECM in neural development.

Transcriptional profiles of murine oligodendrocyte precursor cells across the lifespan.

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Oligodendrocyte progenitor cells (OPCs) are highly dynamic, widely distributed glial cells of the central nervous system (CNS) that are responsible for generating myelinating oligodendrocytes during development. By also generating new oligodendrocytes in the adult CNS, OPCs allow formation of new myelin sheaths in response to environmental and behavioral changes and play a crucial role in regenerating myelin following demyelination (remyelination). However, the rates of OPC proliferation and differentiation decline dramatically with aging, which may impair homeostasis, remyelination, and adaptive myelination during learning. To determine how aging influences OPCs, we generated a novel transgenic mouse line that expresses membrane-anchored EGFP under the endogenous promoter/enhancer of Matrilin-4 (Matn4-mEGFP) and performed high-throughput single-cell RNA sequencing, providing enhanced resolution of transcriptional changes during key transitions from quiescence to proliferation and differentiation across the lifespan. Comparative analysis of OPCs isolated from mice aged 30 to 720 days, revealed that aging induces distinct inflammatory transcriptomic changes in OPCs in different states, including enhanced activation of HIF-1 α and Wnt pathways. Inhibition of these pathways in acutely isolated OPCs from aged animals restored their ability to differentiate, suggesting that this enhanced signaling may contribute to the decreased regenerative potential of OPCs with aging. This Matn4-mEGFP mouse line and single-cell mRNA datasets of cortical OPCs across ages help to define the molecular changes guiding their behavior in various physiological and pathological contexts.

Endoplasmic reticulum stress alters myelin associated protein expression and extracellular vesicle composition in human oligodendrocytes.

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Myelination of the central nervous system is mediated by specialized glial cells called oligodendrocytes (OLs). Multiple sclerosis (MS) is characterized by loss of myelination and subsequent clinical symptoms that can severely impact the quality of life and mobility of those affected by the disease. The major protein components of myelin sheaths are synthesized in the endoplasmic reticulum (ER), and ER stress

has been observed in patients with MS. Extracellular vesicles (EVs) have been shown to carry bioactive cargo and have the potential to be utilized as noninvasive biomarkers for various diseases. In the current study, we sought to determine how ER stress in OLs affected the production of key myelination proteins and EV release and composition. To achieve this, tunicamycin was used to induce ER stress in a human oligodendroglioma cell line and changes in myelination protein expression and markers of autophagy were assessed. EVs were also separated from the conditioned cell culture media through size exclusion chromatography and characterized. Significant reductions in the expression of myelination proteins and alterations to autophagosome formation were observed in cells undergoing ER stress. EVs released from these cells were slightly smaller relative to controls, and had strong expression of LC3B. We also observed significant upregulation of miR-29a-3p in ER stress EVs when compared to controls. Taken together, these data suggest that ER stress negatively impacts production of key myelination proteins and induces cells to release EVs that may function to preemptively activate autophagic pathways in neighboring cells.

MicroRNAs in oligodendrocyte development and remyelination.

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Oligodendrocytes are the glial cells responsible for the formation of myelin around axons of the central nervous system (CNS). Myelin is an insulating layer that allows electrical impulses to transmit quickly and efficiently along neurons. If myelin is damaged, as in chronic demyelinating disorders such as multiple sclerosis (MS), these impulses slow down. Remyelination by oligodendrocytes is often ineffective in MS, in part because of the failure of oligodendrocyte precursor cells (OPCs) to differentiate into mature, myelinating oligodendrocytes. The process of oligodendrocyte differentiation is tightly controlled by several regulatory networks involving transcription factors, intracellular signaling pathways, and extrinsic cues. Understanding the factors that regulate oligodendrocyte development is essential for the discovery of new therapeutic strategies capable of enhancing remyelination. Over the past decade, microRNAs (miRNAs) have emerged as key regulators of oligodendrocyte development, exerting effects on cell specification, proliferation, differentiation, and myelination. This article will review the role of miRNAs on oligodendrocyte biology and discuss their potential as promising therapeutic tools for remyelination.

Microenvironmental regulation of oligodendrocyte replacement and remyelination in spinal cord injury.

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Myelin is a proteolipid sheath enwrapping axons in the nervous system that facilitates signal transduction along the axons. In the central nervous system (CNS), oligodendrocytes are specialized glial cells responsible for myelin formation and maintenance. Following spinal cord injury (SCI), oligodendroglia cell death and myelin damage (demyelination) cause chronic axonal damage and irreparable loss of sensory and motor functions. Accumulating evidence shows that replacement of damaged oligodendrocytes and renewal of myelin (remyelination) are promising approaches to prevent axonal degeneration and restore function following SCI. Neural precursor cells (NPCs) and oligodendrocyte progenitor cells (OPCs) are two main resident cell populations in the spinal cord with innate capacities to foster endogenous oligodendrocyte replacement and remyelination. However, due to the hostile microenvironment of SCI,

the regenerative capacity of these endogenous precursor cells is conspicuously restricted. Activated resident glia, along with infiltrating immune cells, are among the key modulators of secondary injury mechanisms that create a milieu impermissible to oligodendrocyte differentiation and remyelination. Recent studies have uncovered inhibitory roles for astrocyte-associated molecules such as matrix chondroitin sulfate proteoglycans (CSPGs), and a plethora of pro-inflammatory cytokines and neurotoxic factors produced by activated microglia/macrophages. The quality of axonal remyelination is additionally challenged by dysregulation of the supportive growth factors required for maturation of new oligodendrocytes and axo-oligodendrocyte signalling. Careful understanding of factors that modulate the activity of endogenous precursor cells in the injury microenvironment is a key step in developing efficient repair strategies for remyelination and functional recovery following SCI.

Functional myelin in cognition and neurodevelopmental disorders.

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In vertebrates, oligodendrocytes (OLs) are glial cells of the central nervous system (CNS) responsible for the formation of the myelin sheath that surrounds the axons of neurons. The myelin sheath plays a crucial role in the transmission of neuronal information by promoting the rapid saltatory conduction of action potentials and providing neurons with structural and metabolic support. Saltatory conduction, first described in the peripheral nervous system (PNS), is now generally recognized as a universal evolutionary innovation to respond quickly to the environment: myelin helps us think and act fast. Nevertheless, the role of myelin in the central nervous system, especially in the brain, may not be primarily focused on accelerating conduction speed but rather on ensuring precision. Its principal function could be to coordinate various neuronal networks, promoting their synchronization through oscillations (or rhythms) relevant for specific information processing tasks. Interestingly, myelin has been directly involved in different types of cognitive processes relying on brain oscillations, and myelin plasticity is currently considered to be part of the fundamental mechanisms for memory formation and maintenance. However, despite ample evidence showing the involvement of myelin in cognition and neurodevelopmental disorders characterized by cognitive impairments, the link between myelin, brain oscillations, cognition and disease is not yet fully understood. In this review, we aim to highlight what is known and what remains to be explored to understand the role of myelin in high order brain processes.

Astrocytes and Microglia as Major Players of Myelin Production in Normal and Pathological Conditions.

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Myelination is an essential process that consists of the ensheathment of axons by myelin. In the central nervous system (CNS), myelin is synthesized by oligodendrocytes. The proliferation, migration, and differentiation of oligodendrocyte precursor cells constitute a prerequisite before mature oligodendrocytes extend their processes around the axons and progressively generate a multilamellar lipidic sheath. Although myelination is predominately driven by oligodendrocytes, the other glial cells including astrocytes and microglia, also contribute to this process. The present review is an update of the most recent emerging mechanisms involving astrocyte and microglia in myelin production. The contribution of these cells will be first described during developmental myelination that occurs in the

early postnatal period and is critical for the proper development of cognition and behavior. Then, we will report the novel findings regarding the beneficial or deleterious effects of astroglia and microglia, which respectively promote or impair the endogenous capacity of oligodendrocyte progenitor cells (OPCs) to induce spontaneous remyelination after myelin loss. Acute delineation of astrocyte and microglia activities and cross-talk should uncover the way towards novel therapeutic perspectives aimed at recovering proper myelination during development or at breaking down the barriers impeding the regeneration of the damaged myelin that occurs in CNS demyelinating diseases.

Oligodendrocytes and Microglia: Key Players in Myelin Development, Damage and Repair.

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Oligodendrocytes, the myelin-making cells of the CNS, regulate the complex process of myelination under physiological and pathological conditions, significantly aided by other glial cell types such as microglia, the brain-resident, macrophage-like innate immune cells. In this review, we summarize how oligodendrocytes orchestrate myelination, and especially myelin repair after damage, and present novel aspects of oligodendroglial functions. We emphasize the contribution of microglia in the generation and regeneration of myelin by discussing their beneficial and detrimental roles, especially in remyelination, underlining the cellular and molecular components involved. Finally, we present recent findings towards human stem cell-derived preclinical models for the study of microglia in human pathologies and on the role of microbiome on glial cell functions.
